

PATH INTEGRALS:

$$\langle x', t' | x_0, t_0 \rangle = \int_{x_0}^{x'} D[x(t)] e^{iS[x(t)]/\hbar} \rightarrow S[x(t)] = \int_{t_0}^{t_1} dt \mathcal{L}(x, \dot{x})$$

$$\langle x', t' | x_0, t_0 \rangle = \langle x' | \hat{U}(t'-t_0) | x_0 \rangle = \langle x' | e^{-i\hat{H}(t'-t_0)/\hbar} | x_0 \rangle \Rightarrow \langle x' | \psi(t) \rangle = \int dx_0 \langle x', t' | x_0, t_0 \rangle \langle x_0 | \psi(t_0) \rangle$$

Free particle: $\hat{H} = \frac{\hat{p}^2}{2m} \rightarrow$ insert complete $\hat{p}$ b/w $\hat{U}$ and $|x_0\rangle$, operate $\hat{H}$, use $\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$
Evaluate Gaussian integral for p .

Short time intervals: Split $\hat{U}$ into N $\hat{U}$'s of Δt , insert complete $\hat{x}$ between each $\hat{U}$.

for each $\langle x_i | \hat{U} | x_{i-1} \rangle \rightarrow$ Taylor expand $\hat{U}$, insert complete $\hat{p}$ b/w $|x_{i-1}\rangle$ and $\hat{U}$, operate $\hat{H}$, collapse Taylor $\rightarrow$ infinite dx_i and $dp_i \rightarrow$ Gaussian for $dp_i \rightarrow$ gives boxed eqn.

All possible paths contribute to path integral, each scaled by phase $S/\hbar$.

Variation to S_{cl} has no first order contribution ($\delta S = 0$), so for large mass/velocity, the phases for paths in vicinity of cl path add constructively. Non cl path has first order contribution, so phases for nearby paths add destructively. But for micro mass/velocity phases can appear random $\rightarrow$ quantum particles do not follow cl path as phases do not interfere for this to happen.

CENTRAL POTENTIALS:

$$\hat{H} = \frac{\hat{p}_1^2 + \hat{p}_2^2}{2m} + v(|\vec{r}_1 - \vec{r}_2|) = \frac{\hat{P}^2}{2M} + \frac{\hat{p}^2}{2\mu} + v(|\vec{r}|)$$

$\hat{H}$ for two bodies w/ potential b/w them

$$[\vec{r} = \vec{r}_1 - \vec{r}_2, \hat{P} = m_1 \vec{v}_1 + m_2 \vec{v}_2, \hat{p} = \hat{p}_1 + \hat{p}_2]$$

$$\langle \vec{r} | \hat{p} | \psi \rangle = \frac{\hbar}{i} \nabla \langle \vec{r} | \psi \rangle$$

$$\langle \vec{r} | p \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\vec{p} \cdot \vec{r}/\hbar}$$

System is rotationally invariant:

$$[\hat{L}_x, \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2] = 0 \rightarrow \text{potential is invariant.}$$

$\hat{H}$ commutes w/ $\hat{L}_x, \hat{L}_y, \hat{L}_z \therefore \hat{H}, \hat{L}^2, \hat{L}_z$ commute

$$\hat{p} = m_2 \hat{p}_1 - m_1 \hat{p}_2 \xrightarrow{m_1, m_2} \text{in CM frame, } \hat{P} = 0$$

$$\hat{p}^2 = -\hbar^2 \nabla^2$$

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + v(|\vec{r}|)$$

complete set of commuting observables $\rightarrow$ common eigenstates

$$|E, l, m\rangle \rightarrow \hat{H}|E, l, m\rangle = E|E, l, m\rangle; \hat{L}^2|E, l, m\rangle = l(l+1)\hbar^2|E, l, m\rangle; \hat{L}_z|E, l, m\rangle = m\hbar|E, l, m\rangle$$

$$3D \text{ S.E.} \rightarrow \langle \vec{r} | \hat{p}^2 | \psi \rangle + \langle \vec{r} | v(|\vec{r}|) | \psi \rangle = -\frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) \langle \vec{r} | \psi \rangle + \frac{\langle \vec{r} | \hat{L}^2 | \psi \rangle}{2\mu r^2} + v(r) \langle \vec{r} | \psi \rangle = E \langle \vec{r} | \psi \rangle$$

$$\hat{p}_r \rightarrow \frac{\hbar}{i} \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \rightarrow \text{if } |\psi\rangle = |E, l, m\rangle$$

$$\Rightarrow \left[-\frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) + \frac{l(l+1)\hbar^2}{2\mu r^2} + v(r) \right] \langle \vec{r} | E, l, m \rangle = E \langle \vec{r} | E, l, m \rangle \rightarrow \langle \vec{r} | E, l, m \rangle = R(r) \Theta(\theta) \Phi(\phi)$$

$$\rightarrow \text{Radial eqn: } \left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{l(l+1)\hbar^2}{2\mu r^2} + v(r) \right] R(r) = E R(r)$$

$$\rightarrow \hat{L}_z \rightarrow \frac{\hbar}{i} \frac{\partial}{\partial \phi} \Rightarrow \Phi(\phi) \sim e^{im\phi} \Rightarrow m = 0, \pm 1, \pm 2, \dots \Rightarrow l = 0, 1, 2, \dots$$

$$\rightarrow \hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial \phi^2} \right] \langle \theta, \phi | l, m \rangle = Y_{l,m}(\theta, \phi) \Rightarrow \langle \vec{r}, \theta, \phi | E, l, m \rangle = R(r) Y_{l,m}(\theta, \phi)$$

$$\rightarrow \int_0^\infty r^2 dr |R(r)|^2 = 1; \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi |Y_{l,m}(\theta, \phi)|^2 = 1$$

CENTRAL POTENTIAL BOUND STATES:

$$\rightarrow u(r) \text{ behaviour at } r=0: u(r) \xrightarrow{r \rightarrow 0} r^{l+1} \quad Y_{0,0} = \frac{1}{\sqrt{4\pi}}$$

$$\rightarrow \text{Hydrogen: } \hat{H} = \frac{\hat{p}^2}{2\mu} - \frac{Ze^2}{|\vec{r}|} \rightarrow Z=1 \quad \int_0^\infty r^2 dr |R(r)|^2 = \int_0^\infty |u(r)|^2 dr = 1 \quad u(r) = 0$$

$$\rightarrow \text{Radial eqn: } \left[-\frac{\hbar^2}{2\mu} \frac{d^2 u}{dr^2} + \frac{l(l+1)\hbar^2}{2\mu r^2} - \frac{Ze^2}{r} \right] u(r) = E u(r) \quad E \equiv -|E| \quad \rho \equiv \sqrt{\frac{8\mu|E|r}{\hbar^2}} \quad \lambda \equiv \frac{Ze^2}{\hbar} \sqrt{\frac{\mu}{2|E|}}$$

$$\rightarrow \frac{d^2 u}{dr^2} - \frac{l(l+1)}{r^2} u + \left(\frac{2}{\rho} - \frac{\lambda}{r} \right) u = 0 \quad u = A e^{-\rho/2} \quad \text{vanishes at origin except for } l=0$$

$$\rightarrow \text{Large } r \text{ behaviour: } \rho \rightarrow \infty \Rightarrow \frac{d^2 u}{dr^2} - \frac{l(l+1)}{r^2} u = 0 \quad \text{ignore exponential increase term.} \quad u(\rho) = \rho^{\lambda+1} e^{-\rho/2} F(\rho)$$

$$\rightarrow \text{Form of } F(\rho): \frac{d^2 F}{d\rho^2} + \left(\frac{2\lambda+2-1}{\rho} \right) \frac{dF}{d\rho} + \left(\frac{2-\lambda}{\rho} \right) F = 0 \Rightarrow F(\rho) = \sum_{k=0}^\infty c_k \rho^k \rightarrow \text{recursion relation.} \quad \frac{c_{k+1}}{c_k} = \frac{k+1-\lambda}{(k+1)(k+2+2\lambda)}$$

$$\Rightarrow E = -\frac{\mu Z^2 e^4}{2\hbar^2 (1+l+m_r)} \Rightarrow E_n = -\frac{\mu c^2 Z^2 \alpha^2}{2n^2} \rightarrow \alpha = \frac{e^2}{\hbar c} \quad E_n = -13.6 \text{ eV}; a_0 = \frac{\hbar}{\mu c \alpha}$$

$$\rightarrow \text{Hydrogen wavefunctions: } |E, l, m\rangle = |n, l, m\rangle \rightarrow \langle \vec{r} | n, l, m \rangle = R_{n,l}(r) Y_{l,m}(\theta, \phi) = \frac{u_{n,l}(r)}{r} Y_{l,m}(\theta, \phi)$$

$$\rho = \sqrt{\frac{8\mu|E|r}{\hbar^2}} = \frac{2Zr}{na_0}; a_0 = \frac{\hbar}{\mu c \alpha} \Rightarrow u_{1,0} = C_0 e^{-\rho/2} \rightarrow \text{normalize}$$

$$\rightarrow \text{Degeneracy: } 0 \leq l \leq n-1, -l \leq m \leq l \Rightarrow \text{degeneracy} = \sum_{l=0}^{n-1} (2l+1) = n^2 \text{ degenerate.}$$

$$\rightarrow \text{Dirac Delta derivative discontinuity: } \frac{d\psi}{dn} \Big|_{n=0^+} - \frac{d\psi}{dn} \Big|_{n=0^-} = \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^\epsilon \frac{2\mu V(r)}{\hbar^2} \psi(r) dr \quad \text{Green's functions: (Born approx.)}$$

$$\frac{d\psi}{dn} \Big|_{n=0^+} - \frac{d\psi}{dn} \Big|_{n=0^-} = \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^\epsilon \frac{2\mu V(r)}{\hbar^2} \psi(r) dr \quad \text{Sch. Eq: } (\nabla^2 + k^2) \psi(r) = \frac{2\mu}{\hbar^2} V(r) \psi(r) \Rightarrow \psi(r) = A e^{ikz} + \int d\vec{r}' G(r, r') \frac{2\mu V(r')}{\hbar^2} \psi(r')$$

$$\frac{d\psi}{dn} \Big|_{n=0^+} - \frac{d\psi}{dn} \Big|_{n=0^-} = \lim_{\epsilon \rightarrow 0} \int_{-\epsilon}^\epsilon \frac{2\mu V(r)}{\hbar^2} \psi(r) dr \quad (\nabla^2 + k^2) \psi(r) = 0 \Rightarrow \psi(r) = A e^{ikz} \quad (\nabla^2 + k^2) G(r, r') = \delta(r, r') \quad \frac{1}{k^2} \psi(r')$$

Finite spherical well (Deuteron): model of nuclear force b/w $p, n \rightarrow$ spherical well of range a , depth V_0
 Only one bound state $\rightarrow$ all excited states are unbound. $u(\infty) = 0$
 $V(r) = \begin{cases} -V_0 & r < a \\ 0 & r > a \end{cases}$ ground state: $l=0 \Rightarrow -\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} - V_0 u = E u, r < a \Rightarrow \frac{d^2 u}{dr^2} = -\frac{2m}{\hbar^2} (V_0 + E) u = -k^2 u \Rightarrow u = A \sin k r$
 $-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} = E u, r > a \Rightarrow \frac{d^2 u}{dr^2} = -\frac{2m E}{\hbar^2} u = -q^2 u \Rightarrow u = C e^{-q r} \quad u(r \rightarrow \infty) = 0$
 for (II) $\frac{2m V_0 a^2}{\hbar^2} < \left(\frac{\beta_1}{2}\right)^2 \rightarrow 1$ bound state
 u, u' continuous

$\rightarrow$ Infinite spherical well: $V = \begin{cases} 0 & r < a \\ \infty & r > a \end{cases} \quad R(r > a) = 0$
 $r < a: \frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} - \frac{l(l+1)}{r^2} R + k^2 R = 0, \quad k = \sqrt{\frac{2m E}{\hbar^2}}$
 spherical Bessel eqn $\rightarrow$ sph. Bessel funct. $\rightarrow$ sph. Neumann func. $\rightarrow$ not normalizable
 Sph. Bessel func: $j_l(\rho) = (-\rho)^l \left(\frac{1}{\rho} \frac{d}{d\rho}\right)^l \left(\frac{\sin \rho}{\rho}\right)$ B.C. $j_l(ka) = 0$
 normalizing constant A

$\rightarrow$ 3D Isotropic H-0: $V(r) = \frac{1}{2} m \omega^2 r^2$
 $E = (n_x + n_y + n_z + \frac{3}{2}) \hbar \omega = \frac{1}{2} m \omega^2 (x^2 + y^2 + z^2) \Rightarrow \hat{H} = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 r^2 = \hat{H}_x + \hat{H}_y + \hat{H}_z \rightarrow$ all 3 $\hat{H}_i$ commute
 $\Rightarrow$ simultaneous eigenstates $|n_x, n_y, n_z\rangle$
 $\hat{H}_i |n_x, n_y, n_z\rangle = (n_i + \frac{1}{2}) \hbar \omega |n_x, n_y, n_z\rangle$
 $\Rightarrow u = \rho^{\frac{l}{2}} e^{-\frac{1}{2} \rho^2} Y_l^m(\rho)$
 $f(\rho) = \sum_{k=0}^{\infty} c_k \rho^k \rightarrow$ require termination at $k = n_r \Rightarrow E = (2n_r + 2l + \frac{3}{2}) \hbar \omega = (n + \frac{3}{2}) \hbar \omega$
 as $\rho \rightarrow \infty, \frac{d^2 u}{d\rho^2} = \rho^2 u$

PERTURBATION THEORY: $\hat{H} = \hat{H}_0 + \lambda \hat{H}_1 \rightarrow \hat{H}_0 |\varphi_n^{(0)}\rangle = E_n^{(0)} |\varphi_n^{(0)}\rangle \rightarrow |\varphi_n\rangle = |\varphi_n^{(0)}\rangle + \lambda |\varphi_n^{(1)}\rangle + \lambda^2 |\varphi_n^{(2)}\rangle + \dots$

$\rightarrow$ Non-degenerate P.T.: 1st order E shift: $E_n^{(1)} = \langle \varphi_n^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle$
 $E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$

$\rightarrow$ First order eigenstate correction: $|\varphi_n\rangle = |\varphi_n^{(0)}\rangle + \lambda \sum_{k \neq n} |\varphi_k^{(0)}\rangle \frac{\langle \varphi_k^{(0)} | \hat{H}_1 | \varphi_n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} + O(\lambda^2)$
 $\hat{H}_1 = g \hat{S}_z$
 for $\hat{H}_1 = -g \mu_B \hat{S}_z$ states: $|l, m\rangle$
 $\rightarrow$ Degenerate P.T.: N degenerate unperturbed states.

$\rightarrow$ First order E shift: $\sum_{i=1}^N \langle \varphi_{n,i}^{(0)} | \hat{H}_1 | \varphi_{n,i}^{(0)} \rangle c_i = E_n^{(1)} \sum_{i=1}^N S_{ji} c_i$
 $|C\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$
 coefficients for new basis
 Eigenvalue problem: $\hat{H}_1$ is only in basis of degenerate eigenstates.

$\rightarrow$ Stark Effect: Hydrogen in E field $\rightarrow \hat{H}_1 = -\hat{\mu}_e \cdot \vec{E} = e \hat{r} \cdot \vec{E} = e |\vec{E}| z$; $\hat{H}_0 = \frac{p^2}{2m} - \frac{Ze^2}{r} \rightarrow |n, l, m\rangle$

Ground state: First order shift $\rightarrow E_1^{(1)} = e |\vec{E}| \langle 1, 0, 0 | z | 1, 0, 0 \rangle = 0$ ($|n, l, m\rangle$ have parity $(-1)^l$, expect value is 0 if even, odd, even and $\int \text{odd } dr = 0$)
 Second order shift $\rightarrow E_1^{(2)} = \sum_{n \neq 1, l, m} \frac{e^2 |\vec{E}|^2 \langle 1, 0, 0 | z | n, l, m \rangle \langle n, l, m | z | 1, 0, 0 \rangle}{E_1^{(0)} - E_n^{(0)}} \sim E^2$
 1st excited state: $|2, 0, 0\rangle, |2, 1, 0\rangle, |2, 1, 1\rangle, |2, 1, -1\rangle$ basis
 (4-fold deg.)
 $\begin{pmatrix} \langle 2, 0, 0 | \hat{H}_1 | 2, 0, 0 \rangle & \langle 2, 0, 0 | \hat{H}_1 | 2, 1, 0 \rangle & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix} = E_2^{(1)} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{pmatrix}$
 All diagonal terms, terms w/ same l (parity) and terms w/ diff m go to zero
 $\therefore E_2^{(1)} = \{0, 0, 3e |\vec{E}| a_0, -3e |\vec{E}| a_0\} \rightarrow$ eigenstates: $|2, 1, 1\rangle, |2, 1, -1\rangle, \frac{1}{\sqrt{2}}(|2, 0, 0\rangle - |2, 1, 0\rangle), \frac{1}{\sqrt{2}}(|2, 0, 0\rangle + |2, 1, 0\rangle)$
 middle, highest E, lowest E

$\rightarrow$ Relativistic H corrections: $E_n^{(1)} = -\langle n, l, m | \frac{p^4}{8m^3 c^2} | n, l, m \rangle$
 • KE correction: $\hat{H}_K = -\frac{p^4}{8m^3 c^2}$
 $[\hat{H}_K, \hat{H}_0] \rightarrow$ matrix rep.
 of $\hat{H}_K$ in deg. subspace is diagonal, $\Rightarrow E_{n,l} = -\frac{1}{2} m_e c^2 Z^4 \alpha^4 \left[\frac{5}{4n^4} - \frac{1}{n^3} \left(\frac{l+1}{2} + \frac{1}{2l+2} \right) \right]$
 so $\hat{H}_K$ shifts deg. states by same amount $\rightarrow$ can use non deg. PT.

• Darwin term: S-0 for $l=0 \rightarrow \hat{H}_D = \frac{\hbar^2}{8m_e^2 c^2} \nabla^2 \delta(\vec{r})$
 $E_n^{(1)} = \langle n, 0, 0 | \hat{H}_D | n, 0, 0 \rangle = \frac{m_e c^2 Z^4 \alpha^4}{2n^3}$
 not invariant so can use non deg.

$\rightarrow$ Zeeman Effect: $\hat{H}_B = -\hat{\mu} \cdot \vec{B}$
 $= -\left(\frac{-e \hbar}{2m_e c} \hat{L} + \frac{-e \hbar}{2m_e c} \hat{S} \right) \cdot \vec{B}$
 $= \frac{eB}{2m_e c} (\hat{L}_z + \hat{S}_z) = \frac{eB}{2m_e c} (\hat{J}_z + \hat{S}_z)$
 $E_B^{(1)} = eB \langle j, m_j | (\hat{J}_z + \hat{S}_z) | j, m_j \rangle = eB \left(m_j + \frac{1}{2} \right)$
 adding spin of p. we get hyperfine structure (triplet, singlet)

$\hat{J}^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$
 $\hat{J}_z |j, m\rangle = \hbar m |j, m\rangle$
 $\hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$
 $\hat{a} |n\rangle = \sqrt{n} |n-1\rangle$
 $\hat{\lambda} = \frac{\hbar}{2m\omega} (\hat{a}^\dagger \hat{a})$
 $\hat{p} = -i \sqrt{\frac{m\hbar\omega}{2}} (\hat{a} - \hat{a}^\dagger)$
 $\hat{H}_0 = E = (n + \frac{1}{2}) \hbar \omega$
 $(1+x)^r = 1 + rx + \frac{r(r-1)}{2!} x^2 + \dots$
 $\int_0^\infty dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}$

IDENTICAL PARTICLES: exchange op.
 $|a,b\rangle = |a\rangle \otimes |b\rangle \rightarrow \hat{P}_{12}|a,b\rangle = |b,a\rangle = |b\rangle \otimes |a\rangle$
 $|a,a\rangle \rightarrow$ symmetric $\Rightarrow$ fermions cannot be in same state.
 Eigenstates of $\hat{P}_{12}$: $|\psi_s\rangle = \frac{1}{\sqrt{2}}(|a,b\rangle + |b,a\rangle) \rightarrow$ bosons, symmetric
 $|\psi_a\rangle = \frac{1}{\sqrt{2}}(|a,b\rangle - |b,a\rangle) \rightarrow$ fermions, antisymmetric.
 particles cannot be in superposition of $|\psi_s\rangle$ and $|\psi_a\rangle$ - pick one.

Helium Atom: $\hat{H} = \frac{\hat{p}_1^2}{2m_e} + \frac{\hat{p}_2^2}{2m_e} - \frac{Ze^2}{|\vec{r}_1|} - \frac{Ze^2}{|\vec{r}_2|} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$, $Z=2$ (not the best perturbation)
 Eigenstates of $\hat{H}_0$: $\hat{r}_1, \hat{p}_1$ commute w/ $\hat{r}_2, \hat{p}_2$
 Ground state: $|E_0^{(0)}\rangle = |1,0,0\rangle_1 |1,0,0\rangle_2$

$|1,0,0\rangle_1 |1,0,0\rangle_2 \xrightarrow{\hat{P}_{12}} |1,0,0\rangle_2 |1,0,0\rangle_1 = |1,0,0\rangle_1 |1,0,0\rangle_2$
 $\Rightarrow E_{1,1,1,1}^{(0)} = 2(-\frac{1}{2} m_e c^2 Z^2 \alpha^2) = -8(13.6 \text{ eV}) = -108.8 \text{ eV} (Z=2)$
 Hydrogen ground state E

$E_{1,1,1,1}^{(1)} = \langle 1,1,1,1 | \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} | 1,1,1,1 \rangle \rightarrow$ evaluate in pos space
 $|\psi_s\rangle = \int d\vec{r}_1 d\vec{r}_2 |\psi_s(\vec{r}_1, \vec{r}_2)\rangle \langle \psi_s(\vec{r}_1, \vec{r}_2) | \psi_s \rangle \Rightarrow E_{1,1,1,1}^{(1)} = \int d\vec{r}_1 d\vec{r}_2 |\psi_s(\vec{r}_1, \vec{r}_2)|^2 \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$
 From hydrogen, $\langle 1,0,0 | = R_{10} Y_{00} = \frac{1}{\sqrt{\pi}} (\frac{Z}{a_0})^{3/2} e^{-Zr/a_0}$
 $\Rightarrow E_{1,1,1,1}^{(1)} = \frac{5}{8} Z m_e c^2 \alpha^2 = 34.0 \text{ eV}$
 $\Rightarrow E_{1,1,1,1} \approx E_{1,1,1,1}^{(0)} + E_{1,1,1,1}^{(1)} = -74.8 \text{ eV}$
 $E_{1,1,1,1}(\text{exp.}) = -79.0 \text{ eV}$
 Choose z axis for $\vec{r}_1, \vec{r}_2$
 point in $\vec{r}_2$ direction $\Rightarrow \vec{r}_1 \cdot \vec{r}_2 = r_1 r_2 \cos \theta_2$, $|\vec{r}_1 - \vec{r}_2| = (r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2)^{1/2}$
 carry out integration.

Excited states: $\rightarrow 4$ possible states:
 1st exc. st.: $|1,0,0\rangle_1 |2,0,0\rangle_2$
 $E_{1,2,0,0}^{(0)} = -\frac{1}{2} m_e c^2 Z^2 \alpha^2 (\frac{1}{1^2} + \frac{1}{2^2}) = -68.0 \text{ eV}$
 $E^{(1)} = \frac{1}{2} (\langle 1,0,0 | \langle 2,0,0 | \pm \langle 2,0,0 | \langle 1,0,0 |) \times \frac{e^2}{|\vec{r}_1 - \vec{r}_2|} \times (|1,0,0\rangle_1 |2,0,0\rangle_2 \pm |2,0,0\rangle_1 |1,0,0\rangle_2) = J \pm K$
 interacting e^- charge densities
 pure QM term

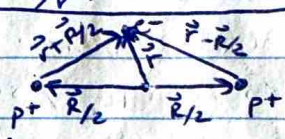
Variational Method: choose trial state $|\psi\rangle$ depending on parameters, minimise $\langle E \rangle = \langle \psi | \hat{H} | \psi \rangle$ for this state to find E_0 as $\langle E \rangle \geq E_0$

Helium: $|\psi\rangle = |1,0,0(\tilde{Z})\rangle_1 |1,0,0(\tilde{Z})\rangle_2$
 (2 Hydrogen ground states w/ nuclear charge $\tilde{Z}$)
 $\hat{H} = \frac{\hat{p}_1^2}{2m_e} - \frac{\tilde{Z}e^2}{|\vec{r}_1|} + \frac{\hat{p}_2^2}{2m_e} - \frac{\tilde{Z}e^2}{|\vec{r}_2|} + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$
 $\langle E \rangle = -m_e c^2 \alpha^2 (\tilde{Z}^2 - 2\tilde{Z}(\tilde{Z}-2) - \frac{5}{8}\tilde{Z})$
 $\langle E \rangle = 2(-\frac{1}{2} m_e c^2 \alpha^2 \tilde{Z}^2) + \frac{e^2}{|\vec{r}_1 - \vec{r}_2|}$
 $\langle E \rangle = 8\tilde{Z}^3 e^2 (\tilde{Z}-2) \langle E \rangle = \frac{5}{8} \tilde{Z} m_e c^2 \alpha^2$
 set $\frac{d\langle E \rangle}{d\tilde{Z}} = 0 \Rightarrow \tilde{Z} = Z - 5/16$ (nuclear charge is screened)
 $\Rightarrow \langle E \rangle|_{\tilde{Z}=Z-5/16} = -77.4 \text{ eV} \rightarrow$ closer to exp.

General variational method:
 choose trial state as exponential or Gaussian, 1) normalize to find N , 2) find $\langle E \rangle$ and minimize to find b
 $\psi = N e^{-br}$ or $N e^{-br^2}$

The Periodic Table: $\hat{H} = \sum_{i=1}^Z \frac{\hat{p}_i^2}{2m_e} - \sum_{i=1}^Z \frac{Ze^2}{|\vec{r}_i|} + \sum_{i < j} \frac{e^2}{|\vec{r}_i - \vec{r}_j|}$ Electrons see screened nuclear charge
 $H: 1s, He: 1s^2, Li: 1s^2 2s$
 $V(r) = \begin{cases} -\frac{Ze^2}{r}, & r \rightarrow 0 \\ -\frac{e^2}{r}, & r \rightarrow \infty \end{cases}$

Covalent Bonding: H_2^+ ion, $\hat{H} = \frac{\hat{p}^2}{2m_e} - \frac{e^2}{|\vec{r} - \vec{R}_1/2|} - \frac{e^2}{|\vec{r} - \vec{R}_2/2|} + \frac{e^2}{R}$
 Trial wavefunctions:
 $\langle r|1\rangle = \frac{1}{\sqrt{\pi a_0}} e^{-|\vec{r} - \vec{R}_1/2|/a_0}$ (e^- in ground state of proton 1)
 $\langle r|2\rangle = \frac{1}{\sqrt{\pi a_0}} e^{-|\vec{r} - \vec{R}_2/2|/a_0}$ (e^- in ground state of proton 2)
 $\langle 1|2\rangle \neq 0$
 $\hat{H}_{11} = \hat{H}_{22} = (E_1 + \frac{e^2}{R}) \cdot \int d^3r \frac{e^2}{R} |\langle r|1\rangle|^2$
 $\hat{H}_{12} = \hat{H}_{21} = (E_1 + \frac{e^2}{R}) \langle 1|2\rangle = \int d^3r \frac{e^2}{R} \langle 1|r\rangle \frac{e^2}{R} \langle r|2\rangle$



Eigenstates of $\hat{H} \rightarrow$ diagonalizing $\Rightarrow E_{\pm} = \frac{1}{1 \pm \langle 1|2\rangle} (H_{11} \pm H_{12}) \rightarrow$ only $|+\rangle$ has a minimum (bonding state), while $|-\rangle$ is antibonding.
 $| \pm \rangle = \frac{1}{\sqrt{2 \pm 2\langle 1|2\rangle}} (|1\rangle \pm |2\rangle)$
 $\langle r|+\rangle$ - even parity
 $\langle r|-\rangle$ - odd parity

SCATTERING

Asymptotic wavefunc: $\psi \rightarrow \frac{Ae^{ikz}}{r} + \frac{Af(\theta, \varphi)e^{ikr}}{r}$
 Differential cross section:

$$\frac{d\sigma}{d\Omega} = |f(\theta, \varphi)|^2$$

Born Approximation
 high beam energy compared to potential.

$$f(\theta, \varphi) = -\frac{u}{2\pi\hbar^2} \int d^3r' v(r') e^{i\vec{q} \cdot \vec{r}'} \quad \vec{q} = \vec{k}_i - \vec{k}_f = \frac{\vec{p}_i - \vec{p}_f}{\hbar}, \quad \vec{k}_i \cdot \vec{r}' = z$$

Prob. current: $\vec{j} = \frac{\hbar}{2mi} (\psi^* \nabla \psi - \psi \nabla \psi^*)$
 from Sch. eqn: $\frac{\partial}{\partial t} (\psi^* \psi) + \nabla \cdot \vec{j} = 0$
 $\frac{d}{dt} \int d^3r \psi^* \psi = - \int d^3r \nabla \cdot \vec{j} = - \oint d\vec{s} \cdot \vec{j}$

$$q^2 = 4k^2 \sin^2 \frac{\theta}{2}$$

if true, then net outflow of prob., so prob. of finding particle in d^3r decreases.

Validity of Born Approx.:

$$|\psi(r')| \sim \psi_{inc} \Rightarrow \left| \frac{\psi_{sc}}{\psi_{inc}} \right| \ll 1 \Rightarrow \left| \frac{\psi_{sc}(0)}{\psi_{inc}(0)} \right| = \left| \frac{2u}{\hbar^2 k} \int_0^\infty dr' e^{ikr'} v(r') \sin kr' \right| \ll 1$$

As $k \rightarrow \infty$ (high E)

$$\Rightarrow \left| \frac{2u}{\hbar^2 k} \int_0^\infty dr' v(r') \sin kr' \right| \sim \frac{uV_0}{\hbar^2 k^2} \ll 1 \Rightarrow \frac{V_0}{E} \ll 1$$

As $ka \rightarrow 0$ (low E),

$$\frac{uV_0 a}{\hbar^2} \ll 1$$

a-ranged potential.

Yukawa potential: $v(r) = g e^{-m_0 r}$

$$f(\theta, \varphi) = -\frac{ug}{2\pi\hbar^2} \int d^3r' \frac{e^{-m_0 r'}}{r'} e^{i\vec{q} \cdot \vec{r}'} \rightarrow \text{choose } z' \text{ parallel to } \vec{q}$$

$$\text{integrating over } d^3r, \quad f = -\frac{2ug}{k^2(m_0^2 + q^2)} \Rightarrow \frac{d\sigma}{d\Omega} = \frac{4u^2 g^2}{\hbar^4 (m_0^2 + 4k^2 \sin^2(\theta/2))^2}$$

$$P_2(\cos\theta) = \sqrt{\frac{4\pi}{24H}} Y_{2,0}$$

Partial wave Expansion: low energy approx.

$$f(\theta) = \sum_{l=0}^{\infty} (2l+1) \frac{e^{i\delta_l} \sin \delta_l}{k} P_l(\cos\theta)$$

$$\psi_{inc} = e^{ikz} = e^{ikr \cos\theta} = \sum_{l=0}^{\infty} i^l (2l+1) j_l(kr) P_l(\cos\theta)$$

$$\Rightarrow \sigma = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l = \sum_{l=0}^{\infty} \sigma_l, \quad \sigma_l = \frac{4\pi}{k^2} (2l+1) \sin^2 \delta_l$$

Hard sphere scattering

$$\sigma_l = \frac{4\pi}{k^2} (2l+1) \sin^2 \delta_l$$

Finite potential well:

$$v(r) = \begin{cases} \infty, & r < a \\ 0, & r > a \end{cases} \Rightarrow \frac{d^2 u}{dr^2} = -k^2 u, \quad k = \sqrt{\frac{2mE}{\hbar^2}}, \quad r > a$$

$$\Rightarrow u = C \sin(kr + \delta_0), \quad r > a$$

$$B.C. \Rightarrow C \sin(ka + \delta_0) = 0 \Rightarrow \delta_0 = -ka$$

$$\Rightarrow \sigma_{l=0} = \frac{4\pi}{k^2} \sin^2 \delta_0 = \frac{4\pi}{k^2} \sin^2 ka \xrightarrow{k \rightarrow \infty} \frac{4\pi a^2}{k^2}$$

$$\xrightarrow{k \rightarrow \infty} \sum_{l=0}^{\infty} \frac{4\pi}{k^2} (2l+1) \sin^2 \delta_l \xrightarrow{ka \gg 1} \sum_{l=0}^{\infty} \frac{2\pi}{k^2} (2l+1) \xrightarrow{avg. = 1/2} 2\pi a^2$$

$$\Rightarrow \frac{d^2 u}{dr^2} = -k_0^2 u, \quad k_0 = \sqrt{\frac{2m(E+V_0)}{\hbar^2}}, \quad r < a$$

$$\frac{d^2 u}{dr^2} = -k^2 u, \quad k = \sqrt{\frac{2mE}{\hbar^2}}, \quad r > a$$

$$\Rightarrow u = A \sin(kr) \quad (u(0) = 0), \quad r < a$$

$$u = C \sin(kr + \delta_0), \quad r > a \Rightarrow ka + \delta_0 \approx \frac{ka}{\tan ka}$$

u, u' continuity, dividing both eqns.

$$\tan(ka + \delta_0) = \frac{ka \tan ka}{ka}$$

$\frac{ka}{\tan ka} \ll 1$, so, if ka is small,

then RHS is small $\Rightarrow$ LHS is small

$$\Rightarrow \delta_0 \approx ka \left(\frac{\tan ka}{ka} - 1 \right)$$

$$\sigma_{l=0} \approx 4\pi a^2 \left(\frac{\tan ka}{ka} - 1 \right)^2$$

TIME DEPENDENT PERTURBATION THEORY: $\hat{H} = \hat{H}_0 + \hat{H}_1(t)$

$$\text{Time } t=0: \text{ Arbitrary state } |\psi(0)\rangle = \sum_n c_n(0) |E_n^{(0)}\rangle \Rightarrow |\psi(t)\rangle = \sum_n c_n(t) e^{-iE_n^{(0)}t/\hbar} |E_n^{(0)}\rangle$$

$$\frac{dc_f(t)}{dt} = \frac{i}{\hbar} \sum_n c_n(t) e^{i(E_f^{(0)} - E_n^{(0)})t/\hbar} \langle E_f^{(0)} | \hat{H}_1(t) | E_n^{(0)} \rangle \rightarrow |E_f^{(0)}\rangle, |E_n^{(0)}\rangle: \text{ orthogonal eigenstates}$$

$$c_n(t) = c_n(0) + \lambda c_n^{(1)} + \lambda^2 c_n^{(2)} + \dots \Rightarrow \frac{dc_f^{(0)}}{dt} = 0$$

$$\text{At } t=0, \text{ system is in } |E_i^{(0)}\rangle \rightarrow \text{eigenstate of } \hat{H}_0, \quad \frac{d}{dt} \langle \psi(t) | \hat{O} | \psi(t) \rangle = \frac{i}{\hbar} \langle \psi(t) | [\hat{H}_1, \hat{O}] | \psi(t) \rangle$$

$$\Rightarrow c_n(0) = \delta_{ni} \Rightarrow c_f^{(0)} = \delta_{fi}, \quad c_n^{(k)}(0) = 0, \quad k > 1$$

$$\Rightarrow c_f(t) = \delta_{fi} - \frac{i}{\hbar} \int_0^t dt' e^{i(E_f^{(0)} - E_i^{(0)})t'/\hbar} \langle E_f^{(0)} | \hat{H}_1(t') | E_i^{(0)} \rangle + \dots$$

$$\Rightarrow \text{Prob. of switching to a diff. eigenstate of } \hat{H}_0, \quad \hat{O}_n(t) = e^{-i\hat{H}_0 t/\hbar} \hat{O}_n e^{i\hat{H}_0 t/\hbar} = \frac{d}{dt} \hat{O}_n = i[\hat{H}_0, \hat{O}_n]$$

H.O. w/ T.D. E field:

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 \quad c_n(\omega) = \frac{i q |E_0|}{\hbar \sqrt{2m\omega}} \int_0^\infty dt' e^{i n \omega t'} e^{-t'/T} \langle n | \hat{x} | 0 \rangle, \quad n \neq 0 \Rightarrow \text{prob. of switching to } n=1: |c_1(\omega)|^2$$

$$\hat{H}_1(t) = -q \hat{x} |E_0| e^{-t/T} \Rightarrow c_n(\omega) = \frac{i q |E_0|}{\hbar \sqrt{2m\omega}} \int_0^\infty dt' e^{i n \omega t'} e^{-t'/T} \langle n | \hat{x} | 0 \rangle = \frac{(q |E_0| T)^2}{2m \hbar \omega} \frac{1}{1 + \omega^2 T^2}$$

$$\text{At } t=0, \text{ system is in } n=0 \text{ state, find prob. that it is in excited state at } t=\infty \Rightarrow \text{only } c_1(\omega) \text{ is non-zero}$$